

New Fashion: Personalized 3D Design with a Single Sketch Input

Tianrun Chen
KOKONI, Moxin (Huzhou) Technology Co., LTD.
China
tianrun.chen@zju.edu.cn

Ling Bai
Huzhou University
China
2022082142@stu.zjhu.edu.cn

Ying Zang*
Huzhou University
China
02750@zjhu.edu.cn

Xinyu Chen
Huzhou University
China
2023388420@stu.zjhu.edu.cn

Shangzhan Zhang
Zhejiang University
China
zhang3z@zju.edu.cn

Wenjun Hu
Huzhou University
China
3180101388@zju.edu.cn

Lingyun Sun
Zhejiang University
China
sunly@zju.edu.cn

Chaotao Ding
KOKONI, Moxin (Huzhou) Technology Co., LTD.
China
2021388117@stu.zjhu.edu.cn

Lanyun Zhu
Singapore University of Technology and Design
Singapore
lanyun_zhu@mymail.sutd.edu.sg

Zejian Li
Zhejiang University
China
zejianlee@zju.edu.cn

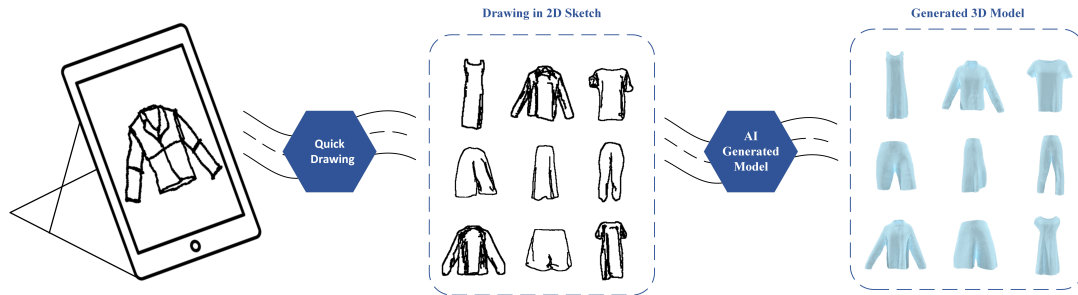


Figure 1: In this paper, we present a novel generative method designed to quickly produce 3D clothing from individual freehand sketches. Even beginners can rapidly create a 3D model easily using our method.

Abstract

In this study, we present a method that enables individuals to generate 3D garment designs from basic sketches, thus making a previously specialized creative field more accessible. Our approach takes a single freehand sketch as input and employs generative deep learning techniques to automatically create high-fidelity 3D garment models using a conditional diffusion 3D generation network.

*Corresponding author.

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Our proposed method comprises two key components: 1) a pre-training phase that uses a 3D prior, drawing on a varied dataset of clothing shapes in 3D to grasp the essential characteristics of these forms, and 2) a sketch-to-prior mapping module that establishes a connection between the sketches and the pre-trained manifold, facilitating the effective generation of 3D shapes. Our method not only overcomes existing limitations but also introduces new possibilities for shape prototyping and exploration in fashion design, thus broadening the scope of creativity for a more diverse audience.

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1 Introduction

Through thoughtfully selected outfits, we express ourselves daily, a privilege once reserved for elite designers but now we want it to be accessible to all.

We tackle the challenge of simplifying 3D design for beginners by using intuitive sketches and leveraging recent generative networks. By adopting a carefully designed conditional diffusion network with a three-stage training strategy and a random sampling augmentation, we maximize the effectiveness of limited data (lack of sufficient labelled data is a major challenge for this task), ensuring quality 3D garment outputs regardless of users' drawing skills (a benefit of conditional generation compared to deterministic regression [Chen et al. 2024]).

Our innovative method surpasses existing benchmarks in quality and fidelity, even accommodating novice users. It seamlessly integrates sketch interpretation and shape generation, ensuring that the resulting 3D models are both high-fidelity and true to the user's vision. This work marks a promising step towards democratizing fashion design, enabling widespread creative contributions and transforming the industry for a new generation of innovators.

2 Method

We adopt a two-stage diffusion framework that utilizes a self-conditioning continuous diffusion model, as in [Zheng et al. 2023]. Specifically, the initial stage generates a low-resolution 3D occupancy volume and the second stage forms a high-resolution sparse volume of SDF. The mesh is extracted from the grids using the Marching Cubes algorithm. Due to the lack of sufficient sketch-3D pair, we designed a three-stage process for training. In Stage 1, "Pre-Training the 3D Priors," we train the model to create 3D shapes from a latent space, utilizing a large-scale pre-trained point cloud encoder, Uni3D [Zhou et al. 2023], which encodes 3D point cloud data (we obtained point cloud data of 3D clothes from Deep3DFashion dataset and cleaned Objaverse dataset) into latent codes within a specific manifold. Note that we use random sampling augmentation for limited 3D data for maximum data exploitation. Stage 2, "Mapping 2D Sketches," focuses on training a specialized 2D sketch encoder that maps sketches to the shared latent space, while keeping the diffusion model unchanged from the first stage. This phase is dedicated to aligning the 2D sketches with the complex, high-dimensional semantic space. In Stage 3, "Joint Fine-Tuning," we simultaneously fine-tune both the 2D sketch encoder and the diffusion model to enhance the spatial and semantic alignment between the shapes and the sketches. This comprehensive approach leverages pre-trained knowledge extensively to improve the quality of the final 3D shapes.

3 Experiments

Our method is evaluated in Deep3DFashion dataset [Zhu et al. 2020] with synthetic sketches and compared with SDFusion [Cheng et al. 2022], LAS-Diffusion [Zheng et al. 2023], Sketch2Model [Zhang et al. 2021], Deep3DSketch [Chen et al. 2023b], and Deep3DSketch+ [Chen et al. 2023a]. The results are shown in Fig. 3 and Tab. 1. We also conduct a user study in which 12 users draw freely using our customized software and get the corresponding 48 3D results. Compared to Deep3DSketch [Chen et al. 2023b], user expressed

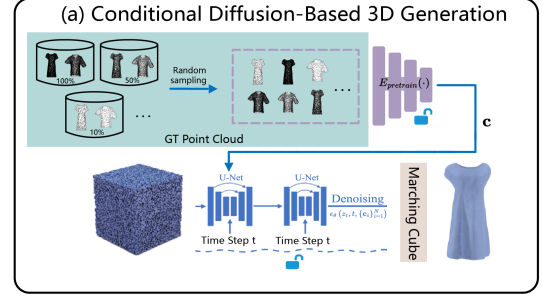


Figure 2: Key Designs of Our Sketch to 3D Garments

Table 1: Quantitative Evaluation of the Synthetic Sketches

Metric	Ours	SDFusion	LAS-Diffusion	Sketch2Model	Deep3DSketch	Deep3DSketch+
CD ↓	0.036	0.056	0.049	0.066	0.038	0.114
FID ↓	42.198	125.111	113.435	131.371	128.857	181.649
MMD ↓	0.076	0.120	0.140	0.115	0.062	0.183
KID ↓	2.470	4.247	4.575	5.619	5.556	6.396

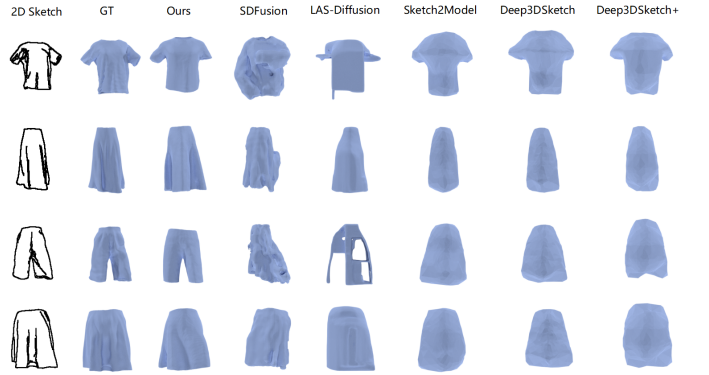


Figure 3: Visualization for Comparisons.

+1.2 more MOS (mean option score) for shape quality and +0.8 score for shape fidelity.

4 Conclusion

This study introduces a method that combines sketching with generative models to democratize 3D clothing creation. Experiments show improved model quality, accuracy, and user satisfaction, with potential to advance 3D fashion design in the metaverse.

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